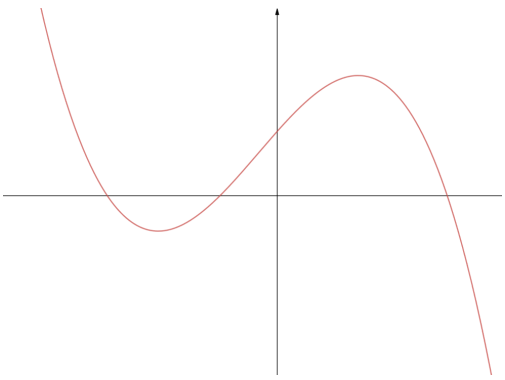
Question	Answer	Marks	Partial Marks
7(d)	54.1 or 54.13… **and** 125.9 or 125.86 to 125.87	**5**	**B4** for 54.1 or 54.13… or 125.9 or 125.86 to 125.87 **M3** for [sin Q =] $\frac{0.5\times82\times55\times\sin76}{0.5\times90\times60}$ oe or **M2** for $0.5 \times 82 \times 55 \times \sin 76 = 0.5 \times 60 \times 90 \times \sin Q$ oe or **M1** for $0.5 \times 82 \times 55 \times \sin 76$ oe or for $0.5 \times 60 \times 90 \sin Q$ = *their* area of *ABC* If **B4** not scored then **SC1** for two angles seen that sum to 180 (from use of sine ratio) but not 0 and 180.
8(a)(i)	(–0.5, 1)	**2**	**B1** for each
8(a)(ii)	$\begin{pmatrix} 7 \\ -3 \end{pmatrix}$	**2**	**B1** for each
8(a)(iii)	7.62 or 7.615 to 7.616	**2**	**FT** *their* **(a)(ii)** **M1** for $(\textit{their } 7)^2 + (\textit{their } -3)^2$ oe
8(a)(iv)	[y =] $-4x - 1$ final answer	**3**	**B2** for answer $-4x + c$ [oe] or for correct equation in different form or for $-4x + -1$ or for $-4m - 1$ OR **M1** for $\frac{-5-7}{1--2}$ oe **M1** for correct substitution shown of (–2, 7) or (1, –5) or *their* (–0.5, 1) into $y = (\textit{their } m)x + c$ oe OR **M1** for $7 = -2m + c$ and $-5 = m + c$ **A1** for $m = -4$ and $c = -1$
8(a)(v)	[y =] $\frac{1}{4}x + \frac{11}{4}$ final answer	**3**	**M1** for grad = $\frac{1}{4}$ oe nfww soi, FT negative reciprocal of *their* gradient from **(iv)** **M1** for correct substitution shown of (5, 4) into $y = (\textit{their } m)x + c$ oe or, if no substitution shown, (5, 4) satisfies *their* final linear equation.

Question	Answer	Marks	Partial Marks
8(b)	$2x^2 + 11x - 21$ [= 0]	**M2**	or **M1** for $8 - 5x = 2x^2 + 6x - 13$ oe or better
	$(2x - 3)(x + 7)$ [= 0] oe or $\frac{-11 \pm \sqrt{11^2 - 4 \times 2 \times (-21)}}{2 \times 2}$ or $-\frac{11}{4} \pm \sqrt{\frac{21}{2} + \left(\frac{11}{4}\right)^2}$ oe	**M2**	Allow correct method to solve *their* quadratic equation e.g. formula, complete the square but not for $2x^2 + 6x - 13$ **M1** FT *their* equation for $2x(x + 7) - 3(x + 7)$ [= 0] or $x(2x - 3) + 7(2x - 3)$ [= 0] or $(2x + a)(x + b)$ [= 0] where $ab = -21$ or $2b + a = 11$ OR **M1** for $\sqrt{11^2 - 4 \times 2 \times -21}$ or for $\frac{-11 + \sqrt{k}}{2 \times 2}$ or $\frac{-11 - \sqrt{k}}{2 \times 2}$ OR **M1** for $\left(x + \frac{11}{4}\right)^2$
	$\left(\frac{3}{2}, \frac{1}{2}\right)$ and (–7, 43)	**B2**	**B1** for one correct pair or for 2 correct x-values or 2 correct y-values
9(a)	Correct sketch of negative cubic crossing the x-axis at –3, –1 and 3 and crossing the y-axis at 9 	**4**	**B1** for any negative cubic shape with two turning points **B2** for three intercepts only with x-axis labelled at – 3, –1 and 3 or **B1** for one or two correctly labelled x-intercepts **B1** for intercept with y-axis labelled at 9 If no graph drawn, **SC1** for all four intercepts labelled on axes.
9(b)(i)	$3 - x + 3x - x^2$ or better or $3 + x + 3x + x^2$ or better or 9 [$- 3x + 3x$] $- x^2$	**M1**	At least 3 of the four terms correct or for the correct expansion of all three brackets with all 8 terms correct
	Correct completion to [y =] $9 + 9x - x^2 - x^3$	**A1**	with no errors or omissions seen